Two Test Plans for Cure Time

AP Statistics · Unit 4 · Topic 4.9 · B: 2027-01-28 · E: 2027-03-17 · TEACHER KEY

CED TETHER

- **Skill 1.E** — Identify an appropriate inference method for significance tests.
- **Skill 1.F** — Identify null and alternative hypotheses.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **EU VAR-7** — The t-distribution may be used to model variation.
- **LO VAR-7.F** — Identify an appropriate selection of a testing method for a difference of two population means. [Skill 1.E]
- **EK VAR-7.F.1** — For a quantitative variable, the appropriate test for a difference of two population means is a two-sample t-test for a difference of two population means.
- **LO VAR-7.G** — Identify the null and alternative hypotheses for a difference of two population means. [Skill 1.F]
- **EK VAR-7.G.1** — The null hypothesis for a two-sample t-test for a difference of two population means, μ_1 and μ_2, is: H_0: μ_1 - μ_2 = 0, or H_0: $\mu_1 = \mu_2$. The alternative hypothesis is H_a: μ_1 - $\mu_2 < 0$, or H_a: μ_1 - $\mu_2 > 0$, or H_a: μ_1 - $\mu_2 \neq 0$, or H_a: $\mu_1 > \mu_2$, or H_a: $\mu_1 < \mu_2$, or H_a: $\mu_1 \neq \mu_2$.
- **LO VAR-7.H** — Verify the conditions for the significance test for the difference of two population means. [Skill 4.C]
- **EK VAR-7.H.1** — In order to make statistical inferences when testing a difference between population means, we must check for independence and that the sampling distribution is approximately normal:
- a. Individual observations should be independent: (i) Data should be collected using simple random samples or a randomized experiment; (ii) When sampling without replacement, check that $n_1 \leq 10\%\ N_1$ and $n_2 \leq 10\%\ N_2$.
- **b. The sampling distribution of $\bar{x}_1$ - $\bar{x}_2$ should be approximately normal (shape): (i) If the observed distribution is skewed, both n_1 and n_2 should be greater than 30; (ii) If the sample size is less than 30, the distribution of the sample data should be free from strong skewness and outliers** — checked for BOTH samples.

Essential question: How do the original question and study design control a defensible two-mean test plan?

DOK-3 skill: 4.C · **FRQ pattern:** `predata-tail-parameter-and-scope`

DOK rationale: Part (c) coordinates the pre-data question, parameter order, sample structure, distribution evidence, and assignment limits to adjudicate two test plans and trace three misleading conclusions.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1 → 3 **Minutes:** 45

Teacher does

- Display the board and ask for one sentence using the study description.
- Begin the 7.8 video follow-along at minute 5.
- Return to the board at minute 33. Students finish the shortened parts and justify their judgment.
- Collect at the door. Score part (c) only using the three required elements.

Students do

- Write a one-sentence first take before the video.

- Complete the video follow-along worksheet.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Which specific number or study detail supports your choice?
- What claim would go beyond the evidence, and why?

Adult role

Circulate and ask for evidence. Accept a concise response; do not require omitted calculations or a second full inference write-up.

[WARN] WATCH FOR

- Choosing a speaker without a reason.
- Copying numbers without explaining what they support.

SCORING PART (c) — E / P / I

Expected elements

- **predata-tail** (required) — Uses the nondirectional original question to justify a two-sided alternative
- **parameters** (required) — Explains why unknown population means belong in the hypotheses
- **cause** (required) — Explains that formula and line differences cannot be separated without random assignment

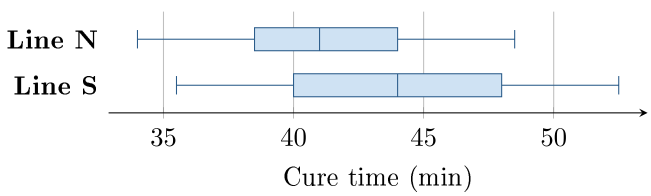
E	Justifies the original two-sided question, population notation, and causal limit.
P	Shows substantive valid reasoning for at least one required element, but does not meet all E requirements. Credit correct reasoning even when another claim is wrong.
I	Shows no substantive valid reasoning for any required element; a name, unsupported choice, or copied number alone does not count.

[STAR] TODAY'S PROBLEM — annotated

Suppose line N uses a new formula for sealant, a material that closes gaps. Line S uses the standard formula. A team takes independent simple random samples of 18 of 600 recent N batches and 18 of 540 recent S batches. Mean cure times are 41.2 and 44.0 minutes; the boxplots are shown. Before sampling, the team asked, "Do the two populations differ in mean cure time?" Formula was not randomly assigned to lines.

Rhea: "Keep the original question and use population means. A difference between lines may have other causes."

Sol: "Since $41.2 < 44.0$, use a lower tail with sample means. A significant result proves the formula caused faster curing."



FIRST TAKE (5 min)

Did the team originally ask whether N was faster or whether the lines differed in either direction? Explain in one sentence.

Key: Ungraded. Everyday language is enough before the video; formal vocabulary belongs in the finish.

Rules callout "HYPOTHESES FOR TWO MEANS (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Define μ_N and μ_S in context.

Key: μ_N and μ_S are the mean cure times, in minutes, for the 600 recent N batches and 540 recent S batches.

DOK 2 **(b)** State H_0 and H_a in N-minus-S order to match the question asked before sampling.

Key: $H_0 : \mu_N - \mu_S = 0$; $H_a : \mu_N - \mu_S \neq 0$.

DOK 3 **(c)** In 3–4 sentences, judge the two plans. Explain why the original question sets the tail, why hypotheses use population means, and why these two lines cannot isolate the formula as the cause.

Key: Rhea is justified because the original question asked about a difference in either direction, so the alternative is two-sided. Choosing a lower tail only after seeing $41.2 < 44.0$ changes that question to favor the observed result. Hypotheses concern unknown population means; the sample means are already known. Because formula was not randomly assigned to lines, other line differences could explain the gap, so a significant result would not establish formula causation.

EXIT

One thing the video changed about my first take: ______________________________